Respiratory Medicine 172 (2020) 106128

Contents lists available at ScienceDirect

Respiratory Medicine

journal homepage: http://www.elsevier.com/locate/rmed

Review article

Anti-muscarinic drugs as preventive treatment of exercise-induced bronchoconstriction (EIB) in children and adults

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ARTICLE INFO

Keywords:
Anti-muscarinic
Exercise-induced bronchoconstriction
Systematic review

ABSTRACT

Regular physical activity is strongly recommended to prevent chronic respiratory diseases, including asthma. On the other hand, vigorous physical training may trigger airway symptoms and bronchoconstriction. The transient airway narrowing occurring because of exercise is named exercise-induced bronchoconstriction (EIB). Despite management according to guidelines, a significant proportion of patients experiences uncontrolled EIB, which thus represents a relevant unmet medical need. In particular, although prevention and treatment of EIB are effectively based on the use of beta-2 bronchodilator drugs, high heterogeneity in individual responses has been reported. Furthermore, even though beta-2 adrenergic drugs remain the mainstay of EIB management, occurrence of tolerance and side effects, as well as doping concerns have been reported with their use. In regard to this, inhaled antimuscarinics could represent an alternative or additional effective and safe bronchodilator therapeutic option for achieving optimal EIB control and minimize adverse events.

The present systematic review aims to collect and provide the most updated and evidence-based literature findings on the efficacy and safety of short- and long-acting inhaled anti-muscarinic drugs for the preventive treatment of EIB in both children and adults.

Take-Home Message: Anti-muscarinic drugs are effective and safe in preventing EIB, despite response variability is reported. Further studies should focus on long-acting molecules, chronic administration and phenotype-driven effects.

1. Introduction

1.1. Physical activity

Regular physical activity is highly recommended by health care systems and scientific guidelines as one of the most effective means to prevent chronic diseases and maintain good health [1,2]. Robust evidence exists regarding the beneficial effect of training and rehabilitation programs in chronic respiratory diseases, including asthma [3]. Indeed, physical activity has been shown to improve pulmonary function, symptoms and quality of life, as well as to reduce airway inflammation and bronchial responsiveness [4–8].

On the other hand, vigorous physical training may trigger respiratory symptoms and airway narrowing by imposing high demands on the respiratory system requiring subjects to ventilate primarily through the mouth and by-pass the nasal filter, with a subsequent increased pulmonary exposure to inhalant allergens, pollutants, irritants and adverse (i.e. cold, dry) environmental conditions [9,10].

1.2. Exercise-induced bronchoconstriction: general

Such phenomenon occurring as a result of exercise is defined as exercise-induced bronchoconstriction (EIB) [11]. Already in the first century AD, Araeteus the Cappadocian described respiratory symptoms induced by physical exercise [12]. However, a scientific objective interest for EIB can be dated back only to 1960, when Jones and

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https://doi.org/10.1016/j.rmed.2020.106128
Received 24 January 2020; Received in revised form 12 August 2020; Accepted 24 August 2020
Available online 28 August 2020

co-workers focused on the transient airway response occurring because of exercise [13].

The prevalence of EIB ranges from 5% to 20% in the general population [14]. EIB has also been reported in almost all asthmatic subjects, reflecting the level of disease control, with EIB occurring more frequently in more severe and uncontrolled patients [15,16]. Furthermore, EIB is particularly frequent, even in the absence of underlying clinical asthma, in athletes [17], children [18] and atopic subjects, as well as following upper respiratory tract infections [19,20].

The intensity, duration and type of training have also been related to the occurrence of airway hyperresponsiveness and symptoms, with EIB being more prevalent in subjects practicing swimming, as well as endurance and winter sport disciplines, such as ice skating, skiing and long-distance running [11].

1.3. Exercise-induced bronchoconstriction: mechanisms

EIB was initially thought to be secondary to a mediator release from mast cells [21]. However, although mediator release does contribute to cause EIB, pathophysiologic changes induced by intense exercising are definitely more complex. Well-designed studies performed by Anderson and co-workers showed that the respiratory water loss and the increase in osmolarity of the airways surface liquid represent major determinants of EIB (osmotic theory) [22]. The vasodilation associated with airways rewarming (thermal theory) has also been reported to play a key-role in inducing bronchial obstruction after exercise [23]. Furthermore, a direct effect on the bronchial epithelium caused by viral infections, occupational agents and exercise itself may represent an alternative mechanism of EIB [24]. Notably, autonomic dysregulation seems also to play a role in causing bronchial obstruction following exercise, both through the basal increased parasympathetic tone and the reflex mechanisms induced by exercise [25,26]. Furthermore, intense physical training may induce a transient status of immune downregulation with a relative shift toward a T2-high response, clinically linked to an increased prevalence of atopy and viral upper respiratory tract infections, both representing relevant risk factors for the onset of bronchial hyperreactivity [27,28].

1.4. Exercise-induced bronchoconstriction: clinical aspects

EIB is clinically characterized by cough, dyspnoea, wheezing or chest tightness and typically develops within 15 min following at least 5–8 min of high intensity (>85% of maximal voluntary ventilation) aerobic training, usually spontaneously resolving within 60 min. However, EIB can also occur during other types of exercise. After an episode of EIB, there is often a refractory period of about 1–3 h in which, if exercise is repeated, the bronchoconstriction is less emphasised [11].

A careful medical history and physical examination is always recommended when suspecting EIB [29]. Questionnaires specifically developed and validated for screening allergic and respiratory diseases in exercisers and athletes, may represent additional useful and easy-to-use diagnostic tools [30,31]. However, studies performed over the past years consistently revealed a poor relationship between the presence of respiratory symptoms and the objective evidence of EIB. Furthermore, pulmonary function tests performed at baseline appear to be poorly predictive of EIB in subjects practicing regular physical activity, often being within the normal values even in the presence of disease [32]. Thus, in order to establish a secure diagnosis of EIB, it is crucial to perform objective testing to confirm dynamic changes in airway calibre and function [11].

Measuring the variation in the forced expiratory volume in the 1 s (FEV_1) before and after a standardized exercise challenge , in the laboratory or in the field, is commonly considered the gold-standard to diagnose EIB [33]. However, there is still open debate about which bronchial provocation challenge should be preferred, and whether a single test could be enough to set a definite diagnosis [34].

1.5. Exercise-induced bronchoconstriction: pharmacological aspects

Several pharmacologic approaches can be adopted to prevent EIB, with bronchodilators representing the mainstay approach [11,35].

Beta-2 adrenergic drugs, both short- and long-acting (SABA and LABA, respectively), when given in a single inhaled dose or with intermittent administration before exercise, are the most effective drugs to prevent EIB [36], providing complete protection against exercise (i.e. FEV_1 fall <10%) in 68% of subjects [37]. The effect usually lasts 2–4 h for SABA and up to 12 h for LABA. However, high heterogeneity has been observed in the efficacy of beta-2 adrenergic agents to prevent EIB depending on the population sample studied, with more variable effects reported in children [38]. Furthermore, the chronic use of SABA and LABA often results in a reduction of the duration and/or magnitude of protection against EIB with cross-reacting tolerance to other beta-2 agonists [37,39]. Importantly, this loss of efficacy seems to be only partially prevented by Inhaled Corticosteroid (ICS) [40]. Daily use of SABA and LABA may also result in EIB worsening [41]. Therefore, SABA and LABA should be used with caution on a daily basis to prevent EIB. In addition, the latest version of the Global Initiative on Asthma (GINA) guidelines no longer recommends SABA-only treatment of asthma in adults or adolescents due to a reported increased risk of asthma-related death and urgent need of asthma-related healthcare [42]. Such evidence may have translational clinical implications also in EIB. At last, special precautions must be taken with respect to the World Anti-Doping Agency (WADA) rules on beta-2 agonists [43] and LABA administration should be always avoided without concomitant use of ICS according to the U.S. Food and Drug Administration (FDA) recommendations [44].

1.6. Exercise-induced bronchoconstriction: anti-muscarinic drugs

The ancient Ayurvedic medicine already used Datura stramonium (a plant with anticholinergic effects) for asthma treatment. With the discovery of atropine, a potent competitive inhibitor of acetylcholine at postganglionic muscarinic receptors, and more importantly with the demonstration of the importance of the parasympathetic nervous system in bronchoconstriction, renewed interest has been placed on the potential value of antimuscarinic agents in asthma and EIB [45]. Vagal innervation is considered as a major determinant of airway tone and represents the reversible component of airflow obstruction [46]. An up-regulated release of acetylcholine (ACh) causes an increased bronchial tone, bronchial hyperresponsiveness and reflex bronchoconstriction, thus leading to the narrowing of the airways.

Bronchoconstriction is primarily regulated by five muscarinic receptors (MR), although to date their exact location and functional role has not been fully clarified. M4R and M5R are mainly implicated in the brain for activating a lot of signalling pathways involved in the modulation of neuronal excitability, synaptic plasticity and feedback regulation of ACh release; M1R, M2R and M3R are expressed in the lung and in the bronchial tree. M1R are mainly distributed in the peripheral lung tissue and in the alveolar walls within parasympathetic ganglia and regulate cholinergic transmission. M2R are found in post-ganglionic nerves where they serve as auto-receptors, on smooth muscle cells (SMC) and on fibroblasts. M3R are predominantly expressed in SMC and mediate SM ACh-induced contraction. In central airways, SM contraction is mediated by vagal innervation, whereas in the peripheral airways the function of M3R is mediated by ACh released in response to inflammatory stimuli. M3R can also be found in sub-mucosal glands where they are responsible for mucus secretion. The action of inhibition of Ach at the MR, by blocking airway smooth muscle contraction, is therefore the main rationale for using anticholinergic drugs in the treatment of EIB. Nonetheless, it has also been suggested that antimuscarinics may exert their beneficial effects through mechanisms other than the reduction of the airway cholinergic tone, some of which still need to be confirmed in humans [47].

Ipratropium bromide (IPB) has been shown to prevent EIB, although the effect is not consistent among patients and may be variable in the same patient over time [48,49].

Furthermore, a large body of evidence has accumulated in recent years to support the use of the long-acting antimuscarinic tiotropium bromide in asthma [50,51], and its use is now recommended by current international GINA guidelines for chronic treatment of patients >12 years old with most severe and frequently exacerbated forms [42].

1.7. Unmet needs and study aims

The efficacy and safety of anticholinergic drugs as a prophylactic treatment of EIB have never undergone any comprehensive review and accurate assessment until now. The current paper therefore aims at providing a state of the art and evidence-based systematic review of the role of antimuscarinic agents in the EIB prevention in children and adults.

Current practice guidelines and parameters on EIB recommend to distinguish between the occurrence of bronchial obstruction after exercise in asthmatic patients (EIB with asthma - EIBa) and the occurrence of EIB in subjects without other symptoms and signs of underlying clinical asthma (EIB without asthma - EIBwa) [11]. Despite this novel and perhaps more appropriate nomenclature, all the studies considered for the present review were published before these terms were proposed and therefore do not provide sufficient information to differentiate between the two conditions. Thus, it has been decided to use the more general term EIB throughout the text for better consistency and clarity.

2. Methods

2.1. Electronic searches

Electronic searches were undertaken in Medline, Scopus, Isi Web of Knowledge and The Cochrane Central Register of Controlled Trials (CENTRAL) from the date of inception to 2019, using the following keywords: (asthma OR bronchoconstriction) AND (exercise OR "physical activity") AND (anticholinergic OR antimuscarinic OR bronchodilator OR tiotropium OR ipratropium OR oxitropium OR aclidinium OR glycopyrronium). No language or publication restrictions were applied to the search.

2.2. Searching other resources

Reference lists of included studies and recent textbooks were searched for additional potentially relevant papers. National and international clinical trial websites (www.clinicalstudyresults.org; www.clinicaltrials.gov; www.fda.gov; www.eudract.ema.europa.eu) were also checked for ongoing and unpublished studies.

2.3. Inclusion criteria

2.3.1. Types of studies

All original study types (i.e. randomised control trials, case-control studies, cohort studies), of any study design (i.e. parallel groups, cross over), either blinded or open-label, published in full-text and primarily aimed at assessing the preventive effects of inhaled anti-muscarinic drugs on EIB were considered eligible for inclusion in the present review. Data published in abstract form only were excluded.

2.3.2. Types of participants

Both children and adults with a clear clinical history of EIB and/or a positive response to any standardized bronchoprovocation challenge recognised by the International Olympic Committee (IOC) as a valid test to diagnose EIB were considered eligible for inclusion [52].

2.3.3. Types of interventions

Any dose of an antimuscarinic drug (both short- and long-acting) administered by inhalation as a preventive treatment before a bronchoprovocation challenge and within a time-period not exceeding their pharmacological half-life (i.e. 3 h for SAMA and 24 h for LAMA).

2.3.4. Types of comparisons

Placebo or alternative active bronchodilator treatments (i.e. beta-2 agonists) administered by inhalation as a preventive treatment before a bronchoprovocation challenge and within a time-period not exceeding their pharmacological half-life.

2.3.5. Types of outcome measures

2.3.5.1. Primary outcomes.

- Mean % fall in FEV_1 [= 100 x (baseline pre-exercise value - lowest post-exercise value)/baseline pre-exercise value]

- Mean % protection [= 100 x (max % fall FEV_1 placebo - max % fall FEV_1 anti-muscarinic drugs)/max % fall FEV_1 placebo]

- Side-adverse effects.

2.3.5.2. Secondary outcomes.

- Number of subjects with a % fall in FEV_1 >10%

- Mean % fall in other pulmonary function parameters (i.e. PEFR, FEF 25%–75%, MEF 50%)

- Onset of tolerance (in studies evaluating chronic drug administration).

2.4. Selection of studies

Titles and abstracts of papers identified through the search strategy were reviewed independently by two authors, and articles potentially fulfilling the inclusion criteria were retrieved in full text. From full texts, the same two authors independently established whether those studies actually met the inclusion criteria. Studies that did not fulfil all the inclusion criteria were excluded, and reasons for exclusion were reported. The percentage of agreement was recorded using the Cohen's kappa coefficient (κ) which is a statistical measure of concordance between two raters. Any disagreement was solved by consensus. If the two review authors did not reach an agreement, a third review author adjudication was adopted to solve disagreements. Review authors were not blinded to authors, journals, results, etc.

2.5. Data extraction and management

Data extraction was performed independently by two review authors. Full texts were screened, and bibliographic details, as well as data regarding study design, participants, intervention, comparison and outcomes, were recorded in predefined forms. All data, numerical calculations and graphic extrapolations were independently confirmed. We did not deal with missing data.

2.6. Assessment of risk of bias in included studies

We assessed the risk of bias in included studies as high, low or unclear using The Cochrane Collaboration 'Risk of Bias' tool (Handbook 5.1) [53] and the following headings:

- Random sequence generation and allocation concealment (selection bias).
- Blinding of participants and personnel (performance bias).

- Blinding of outcome assessment (detection bias).
- Incomplete outcome data (attrition bias).
- Selective reporting (reporting bias).
- Other bias.

Discussion or third-party adjudication was used to resolve disagreements only three times.

2.7. Rating overall confidence in the results of the review

The AMSTAR-2 critical appraisal tool for systematic reviews including randomised and/or non-randomised studies of healthcare interventions has been adopted for rating the overall confidence in the results presented [54].

3. Results

3.1. Results of the search

The search strategy yielded 3533 articles of which 3518 were identified through electronic database searching (PubMed 1214; ISI Web of Knowledge 480; Scopus 1410; CENTRAL 414) and 15 were identified through searching of other resources (Fig. 1). After having removed duplicates and screened remaining titles and abstracts, forty-six articles were considered potentially eligible for inclusion and were retrieved in full text.

Among these, thirty-four studies actually met all predefined criteria and were therefore included in the present review, while 12 were excluded with reasons (Fig. 1). Cohen's kappa coefficient for study selection was >0.8 supporting a near perfect agreement between the two authors (as mentioned, the two reviewers who selected the studies to be included did not reach an agreement in only three cases with the need of a third review author adjudication).

Main characteristics of included studies are summarized in Table 1.

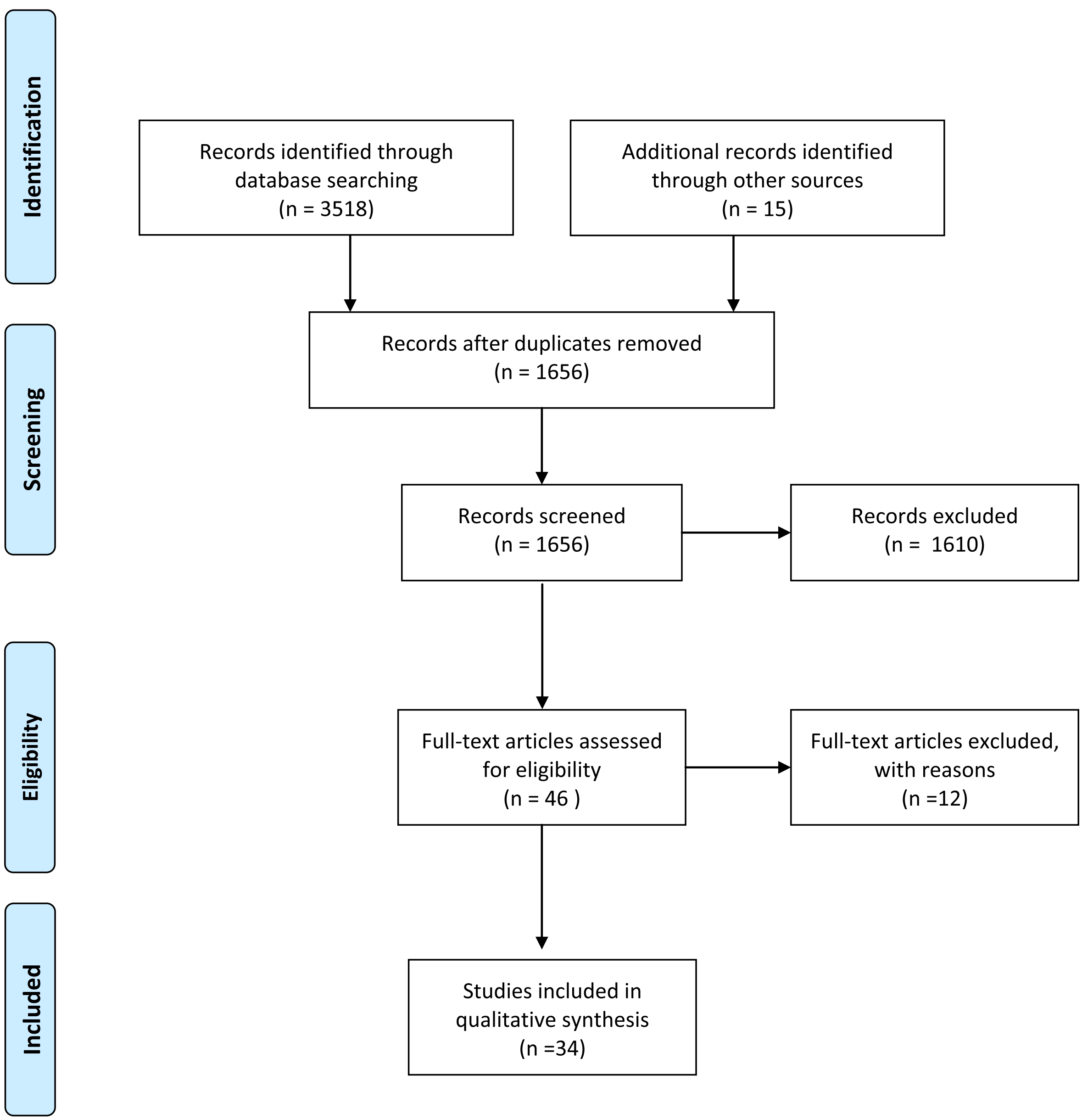

Fig. 1. Search strategy flow-chart.

Table 1
Main characteristics of included studies.

Characteristics		#
Study type	RCT	28
	Non-RCT	6
Blinding	Double-Blind	21
	Single-Blind	6
	Not mentioned	7
Study design	Cross-over	33
	Parallel	1
Population	Adults	14
	Children	11
	Both	9
Intervention[a]	Atropine	6
	Ipratropium bromide	25
	Oxitropium	2
	Combination therapy	2
Comparator	Placebo	31
	Active	3
Outcomes[a]	Primary	29
	Secondary	25
Administration	Acute	34
	Chronic	0
Molecule	SAMA	34
	LAMA	0
Bronchoprovocation test	Exercise	31
	Eucapnic Voluntary Hyperpnoea	2
	Exercise + Hyperosmolar test	1
Year of publication	$<$1970	1
	1970–2000	31
	$>$2000	2

[a] Some interventions and outcomes of interest were considered in the same studies.

Twenty-eight studies were randomised controlled trials. Among them, twenty-one were double-blinded. Thirty-three were performed according to a crossover design while 1 was run in parallel. Adult subjects were studied in 14 studies and children in 11, while the remaining 9 studies included both paediatric and adult populations. The effects of atropine, IPB and oxitropium were evaluated in 6, 25 and 2 study respectively, while those of a combination therapy (i.e. oxitropium + ipratropium and atropine + ipratropium) were assessed in 2 studies. Three studies considered an active comparator, while 31 were placebo controlled. All studies assessed the effects of a single drug dose administered before a bronchoprovocation challenge (acute administration). Most of the studies (31) adopted the exercise challenge as bronchoprovocation test, 2 the Eucapnic Voluntary Hyperpnoea (EVH) test and 1 the hyperosmolar test in combination with exercise. Interestingly, only 2 papers were published after the year 2000 and no one addressed the effects of long-acting muscarinic agents. All the included studies were judged at unclear risk for one or more considered bias headings [53]. None of the included studies was powered for the considered primary outcome(s). Having the systematic review no critical weakness according to the AMSTAR-2 rating, the results can be considered accurate, comprehensive and confident.

3.2. Included studies

Included studies are synthetically listed in Table 2. In view of their higher quality of evidence, only the 28 randomised controlled trials below are discussed in detail.

Eighteen adult asthmatics took part in a double-blind crossover study comparing the effect on EIB of pre-treatment with aerosolized disodium cromoglycate (DSCG), IPB, fenoterol, DSCG + IPB and saline. EIB was fully prevented by using the β2-agonist fenoterol, while the combination DSCG + IPB reduced the post-exercise bronchoconstriction more than any of the two drugs administered separately [55].

A standardized treadmill running in seven atopic adult asthmatics was performed after the administration of placebo, oral ketotifen, inhaled clemastine, IPB and DSCG, in a random single blind-fashion on different days. The mean (±SE) post-exercise percentage fall in FEV_1 was 47 ± 6.95%, 39 ± 8.35%, 27 ± 7.17%, 23 ± 7.69% and 70 ± 4.62%, respectively. The study showed a significant protection against EIB with IPB in four out of seven patients [56].

In a double-blind randomised trial, 9 patients performed an exercise challenge test after 80 mcg of IPB or placebo. The mean specific airway resistance (SRaw) with IPB decreased from 7.8 to 5.1 cmH_2O per litre per sec ($p < 0.01$), even if the individual responses varied largely with a mean maximum percent increase in SRaw of 231% after placebo and 173% after IPB ($p < 0.05$) [57].

Twelve children with asthma were challenged with methacholine and exercise after inhalation of 125, 250, 500, and 750 mcg of IPB compared with saline. A small and similar bronchodilation was observed 60 min after the administration of each dose of IPB. Mean percent fall in FEV_1 after exercise was 36.8%, 18.3%, 23.7%, 27.1% and 23.2% following inhalation of saline or 125, 250, 500 and 750 mcg of IPB, respectively. No side effects were recorded during the study [58].

In a placebo-controlled, double-blind, cross-over study, 21 subjects performed 4 exercise challenges at different times during the day. The bronchodilator effect of IPB was +13.11 ± 10.99% in the morning ($p = 0.001$ compared to baseline values), and +7.25 ± 11.37% in the evening ($p > 0.05$). In the morning, the effect on EIB was 0.58 ± 0.29 L on the placebo day and 0.38 ± 0.22 L on the treatment day ($p = 0.01$), while no difference was observed between the two interventions in the evening. The Authors therefore suggested that the use of IPB for the treatment of EIB should take into consideration the time of administration[49] [].

The effect of IPB and DSCG were studied in a randomised, cross over, placebo control, single blind trial. In six out of the nine subjects enrolled, DSCG and IPB were both effective in preventing EIB. In two patients IPB was effective while DSCG gave no protection. In the remaining patient DSCG was more effective than IPB [59].

EIB was provoked by standardized treadmill running for 6 min in 15 asthmatic children, aged 5–14 years, after the administration, on different days, of placebo, salbutamol, sodium cromoglycate, choline theophyllinate and atropine methonitrate aerosol in a randomised fashion. The mean post-exercise percent fall in peak expiratory flow rate was 45.2%, 4.1%, 19.6%, 18.3% and 24.9%, respectively. The proportion of children having significant improvement of their EIB compared with those taking the placebo was 100% for salbutamol, 80% for cromoglycate and theophyllinate, and 60% for atropine [60].

The effect of 0.5 mg nebulized IPB on the cardio-respiratory response to a progressive maximal cycle exercise test was compared with a matched placebo in eight mild asthmatic and eight healthy subjects. The FEV_1 in the asthmatic group was significantly ($p < 0.02$) higher after IPB compared to placebo. However, since the percentage fall in FEV_1 after exercise was not significantly reduced by IPB vs placebo, the improvement in the post-exercise FEV_1 was referred by the authors to the pre-exercise bronchodilatation [61].

The effect of 490 mcg of IPB on EIB was compared with 400 mcg of fenoterol and placebo in a single blind, randomised, cross-over trial. Seventeen children between 9 and 18 years were enrolled. IPB was shown to be an effective bronchodilator comparable to, but not better than, fenoterol. Both active drugs used in this study were well tolerated and had minimal side effects. Slight tremor was noted more frequently with fenoterol than IPB. Other reported side effects such as gastric irritation, blurred vision, headache did not significantly differ [62].

Sanguinetti et al. studied 12 asthmatic patients with EIB performing an exercise challenge after the administration of fenoterol 400 or 200 mcg, or fenoterol 200 mcg plus IPB 80 mcg, or placebo in a randomised order. The results show that, at 5 h after the administration of each drug, the protective action against EIB was only partial [63].

The anticholinergic effect of IPB was compared with placebo to investigate effects on post-exertional wheezing, pulmonary function and respiratory sound analysis. IPB did not abolish EIB, but improved pre-exercise lung function. Wheezing after IPB was significantly reduced, later in onset and mainly expiratory at 20 min after exercise [64].

Table 2
List of included studies.

Author	Year	Study Design	Population	Administration	Anticholinergic	End-points	Bronchoprovocation challenge
Bonaventura	2010	RCT, double blind, cross-over, placebo control	Adults	Acute	IPB	FEV1	Exercise
Boner	1987	RCT, double blind, cross-over, placebo control	Children	Acute	IPB	FEV1	Exercise
Boner	1989	RCT, cross-over, placebo control	Children	Acute	IPB	FEV1	Exercise
Boulet	1989	RCT, cross-over, double blind, placebo control	Adults	Acute	IPB	FEV1, FVC	Exercise + Hyperosmolar
Borut	1977	RCT, cross over, double blind, placebo control	Children	Acute	Atropine, IPB	SGaw, FVC, FEV1, FEF	Exercise
Bratteby	1986	RCT, cross-over, double blind, placebo control	Children	Acute	IPB	FEV1, MEF, FEF	Exercise
Bungaard	1980	RCT, cross-over, double-blind, placebo control	Both	Acute	IPB	PEF	Exercise
Chan Yeung	1977	RCT, cross-over, single blind, placebo control	Adults	Acute	IPB	FEV1	Exercise
Dorward	1982	RCT, cross-over, single blind, placebo control	Both	Acute	IPB	FEV1	Exercise
Finnerty	1993	RCT, cross-over, double blind, placebo control	Adults	Acute	IPB	FEV1	Exercise
Freeman	1992	RCT, cross-over, double blind, placebo control	Adults	Acute	IPB	FEV1, VO2 max	Exercise
Godfrey	1976	RCT, cross-over, placebo control	Children	Acute	Atropine	PEF	Exercise
Hartley	1980	RCT, cross-over, single blind, placebo control	Adults	Acute	IPB	FEV1, PEFR	Exercise
Johnson	1984	RCT, cross-over, double blind, placebo control	Adults	Acute	Atropine	FEV1, sGAW	Exercise
Knopfli	1999	RCT, cross-over, double blind, placebo control	Adults	Acute	IPB	FEV1, MEF, FVC	Exercise
Knopfli	2004	RCT, cross-over, double blind, placebo control	Children	Acute	IPB	FEV1	Exercise
Larsson	1982	RCT, cross over, double blind, placebo control	Adults	Acute	Oxitropium, IPB	FEV1, VC	Exercise
Magnussen	1992	RCT, parallel group, double blind, placebo control	Adults	Acute	IPB	sRaw	Exercise
Pasterkamp	1985	RCT, cross-over, double blind, placebo control	Both	Acute	IPB	FEV1, Vmax, sGAW, post-exertional wheezing	Exercise
Poppius	1986	RCT, cross-over, double blind, placebo control	Adults	Acute	IPB	FEV1, sGAW	Exercise
Sanguinetti	1986	RCT, cross-over, double blind, placebo control	Both	Acute	IPB	TGV, FEV1, sGAW, PEF, Vmax50, Vmax75	Exercise
Tashkin	1977	RCT, cross-over, double blind, placebo control	Children	Acute	Atropine	FVC, FEV1, FEF, sGAW	Exercise
Taytard	1987	RCT, cross-over, double blind, placebo control	Both	Acute	Oxitropium	FEV1, MMEFR, FVC, Raw	Exercise
Thomson	1978	RCT, cross-over, double blind, placebo control	Both	Acute	IPB	FVC, FEV1, MMEF	Exercise
Tullet	1982	RCT, cross over, single blind, placebo control	Both	Acute	IPB	FEV1, MMFR	Exercise
Wilson	1984	RCT, cross over, double blind, placebo control	Children	Acute	IPB	FEV1, PD20	Eucapnic hyperventilation
Wolkove	1981	RCT, cross-over, double blind, placebo control	Adults	Acute	IPB	FEV1, MMFR	Exercise
Yeung	1980	RCT, cross-over, single blind, placebo control	Both	Acute	IPB	PEFR, FEV1, FEF	Exercise
Beil	1978	Cross over, active comparator	Adults	Acute	Atropine	TGV, Raw	Exercise
Jones	1963	Cross over, active comparator	Children	Acute	Atropine	FEV1	Exercise
Russo	1986	Cross-over, placebo control	Children	Acute	IPB	FVC, FEV1, FMF	Exercise
Smith	1988	Cross over, placebo control	Both	Acute	IPB	FEV1	Eucapnic hyperventilation
Svenonious	1988	Cross over, single blind, placebo control	Children	Acute	IPB	FEV1, TGV	Exercise
Zielinski	1977	Cross over, active comparator	Adults	Acute	Atropine	PEF	Exercise

The effect of oxitropium bromide (OTB) has been studied in ten subjects with documented EIB. While there was no change after inhalation of placebo, administration of OTB led to bronchodilatation and totally blocked post-exercise bronchoconstriction in 7 patients, and partly did so in 2 [65].

Eucapnic hyperventilation dose response curves were constructed for 11 asthmatic children 8–16 years before and after pre-treatment with placebo or IPB. Complete protection was achieved by ipratropium bromide 40 mcg in six children and by 200 mcg or more in a further four. The remaining child was unaffected by any dose of IPB up to 1500 mcg [66].

The preventive effects of inhaled salbutamol, IPB and DSCG on bronchospasm induced by exercise or hyperosmolar saline aerosol was tested in 11 asthmatic patients. The three drugs protected against the two types of challenges in almost all subjects. Although the authors observed a large interindividual variability, there was no statistically significant difference in the mean percent protection offered by salbutamol, IPB or DSCG [67]. Borut et al. evaluated the effectiveness of different doses of nebulized IPB (40 mcg and 80 mcg) and aerosolized atropine sulfate (1 mg) in 20 children (mean age 13.1 ± 2.15 years) with a randomised, cross-over double-blind protocol. After pre-treatment with IPB or atropine, exercise caused specific airway conductance (SGaw) to fall to values that were not significantly different from pre-treatment medication values, but were significantly higher than values following exercise without pre-treatment or after pre-treatment with placebo. Side effects encountered during the study were minimal. There were four episodes of thirst in 3 patients; 2 of the episodes occurred with IPB 40 mcg, 1 with IPB 80 mcg, and 1 with placebo. One patient reported blurred vision on two separate occasions. One patient reported increased fatigue after taking IPB 40 mcg [68].

Eight patients with EIB participated in a single-blind, placebo control trial performing an exercise challenge after inhalation of lignocaine 48 mg, sodium cromoglycate 12 mg, and IPB 120 μg. The post-exercise maximal percentage fall in FEV_1 (mean ± SE) after saline, lignocaine, sodium cromoglycate, and IPB were 38.1 ± 5.0%, 34.5 ± 6.1%, 11.3 ± 3.7% and 19.3 ± 7.4%, respectively. The inhibitory effects of DSCG and IPB were significant whereas lignocaine failed to show a bronchoprotective effect [69].

The effects of placebo, sodium cromoglycate, IPB, and IPB plus sodium cromoglycate were analysed in thirteen patients in a random double-blind fashion. Sodium cromoglycate ($p < 0.02$), IPB ($p < 0.01$) and IPB plus sodium cromoglycate ($p < 0.01$) all significantly inhibited the percentage fall in FEV_1 after exercise in responder subjects (i.e. those with a baseline ratio of VC breathing air and He-O_2:$V_{50He}/V_{50air} > 1.20$) [70]. In addition, no side effects were noted during the study after any of the drugs used.

An exercise challenge was performed to compare the efficacy of IPB, 0.1, or 1 mg against placebo. The mean fall in FEV_1 was 22.3%, 19.5% and 12.5% with placebo, 0.1 mg and 1 mg IPB, respectively. Only the higher dose of IPB had a significantly better effect compared to placebo. Similar results were obtained using PEFR and no side effects were noted during the study [71].

Ten asthmatic patients were enrolled in a randomised double-blind study, to perform an exercise test after IPB 0.5 mg, terfenadine 180 mcg, and placebo. The mean maximum percentage fall in FEV_1 after exercise was 33.6 ± 3.8% after placebo, 27.2 ± 3.6% after terfenadine, 21.8 ± 3.9% after IPB and 15.1 ± 4.1% after the combination of both drugs [72].

Poppius et al. evaluated 10 asthmatic adults through four treadmill exercise tests before the administration of placebo or IPB at the dose of 0.2, 1 and 2 mcg. The mean post exercise fall from the post-treatment pre-exercise values in the four groups was 14%, 14%, 16% and 16% for FEV_1 and 59%, 52%, 52% and 55% for SGaw, respectively. Regarding side-effects, dryness of the mouth was reported by six patients after 2 mcg of IPB and by three after 1 mcg of IPB [73].

In a randomised double-blind crossover placebo control trial, 15 asthmatic children, 9–15 years, were enrolled to perform exercise tests without and after the administration of isoproterenol, atropine, DSCG and both atropine and isoproterenol. The treatment with isoproterenol or atropine showed a larger post-exercise response in terms of specific conductance than that following DSCG [74].

Bratteby et al. demonstrated a better protection on EIB of a combined therapy (IPB – beta-2 agonist) rather than salbutamol alone among 13 asthmatic patients and no side effects were reported [75].

In another study, IPB showed a more beneficial effect on EIB compared with no treatment and placebo. Moreover, the difference in FEV_1% drop following exercise- and cold-induced challenge between inhaled IPB and no treatment was positively correlated to vagal activity ($r = 0.91$, $p = 0.002$), as well as the difference in FEV_1% drop following exercise- and cold-induced challenge between inhaled IPB and placebo ($r = 0.90$, $p = 0.002$) [76].

Six well trained healthy adult runners performed exercise provocation at different temperatures before and after the administration of IPB that caused an increase in FEV_1 versus placebo [77]. In this trial OTP was compared, in terms of EIB prevention, with IPB, fenoterol and placebo in 8 patients performing an exercise test. Only 4 patients had benefits from IPB and 1 from OTP [78].

Glycopyrrolate was compared to atropine and placebo on FEV_1 and sGaw before and after exercise test. The post-exercise measurements were significantly higher after atropine and glycopirrolate compared to after placebo. Systemic side effects were minimal with Glycopyrrolate compared to atropine [79].

In a double-blind placebo control RCT, IPB was administered to 15 children, 10–15 years, in comparison to sodium cromoglycate and verapamil. After exercise test, IPB and sodium cromoglycate showed a significant better response in terms of percentage fall in FEV_1 versus placebo and verapamil. No side effects were recorded during the study [80].

At last, in the double blind, placebo control, cross-over study of Wolkove and co-authors, IPB didn't show a valid protection against EIB compared with placebo and no side effects were recorded [81].

3.3. Excluded studies

Twelve articles were excluded for the following reasons: 3 papers were duplicates of included studies [82–84], 2 articles were non-original studies [85,86], 2 manuscripts were retracted [87,88] and 5 didn't focus on EIB [89–93].

4. Discussion

Collected literature findings supports an effective role of anti-muscarin drugs in the preventive treatment of EIB in both children and adults.

Existing evidence appears to be of moderate-high quality, being based in most cases on studies with robust design (i.e. 28/34 RCT) and addressing at least one of the considered primary outcomes (i.e. 29 trials). Furthermore, the lack of any significant difference, compared to placebo or alternative active treatments, in the number of severe adverse reactions and side-effects supports a fully harmless profile for these drugs, despite safety has never been investigated as a primary outcome.

However, it must be highlighted that, relevantly, all the included studies were not powered in terms of sample size and were judged at unclear risk of bias for one or more considered headings. This may be explained by the high number of papers conducted before the year 2000, when robust statistical methodologies and criteria for reporting risks of bias were less common.

Moreover, almost no study evaluated the effects of long-acting molecules (i.e. tiotropium, aclidinium, glycopyrronium), regretfully preventing to draw final conclusions on these potentially extremely beneficial therapeutic options against EIB, particularly for exercisers practicing endurance sport disciplines.

Furthermore, all the included papers addressed acute drug administrations with therefore no available data on the chronic preventive effects of anticholinergic drugs on EIB. These would be highly important for assessing the reported potential onset of tolerance.

Also, with only 2 studies published after the year 2000 it is not possible to discriminate between any possible difference in the preventive action against EIB with- and without asthma given that such distinction, recommended by the latest guidelines on EIB, has been only recently suggested.

However, it is to point out that, with all the indicated limitations taken into account, there is a useful clinical content emerging from the studies on anti-muscarinic drugs as preventive treatment of EIB in children and adults. Fig. 2 reports the mean % fall in FEV_1 after exercise for each treatment for 19 studies in which data were available. Furthermore, the % Protection was computed for each treatment as $Mean\%protection = \frac{(max\%fallFEV1placebo - max\%fallFEV1anti-muscarinicdrugs)}{max\%fallFEV1placebo} x100$ and averaged among studies. The mean % protection for IPB was 33.5% and the difference of mean % fall in FEV_1 between IPB and Placebo was statistically significant ($p = 0.001$). The mean % protection for Atropine

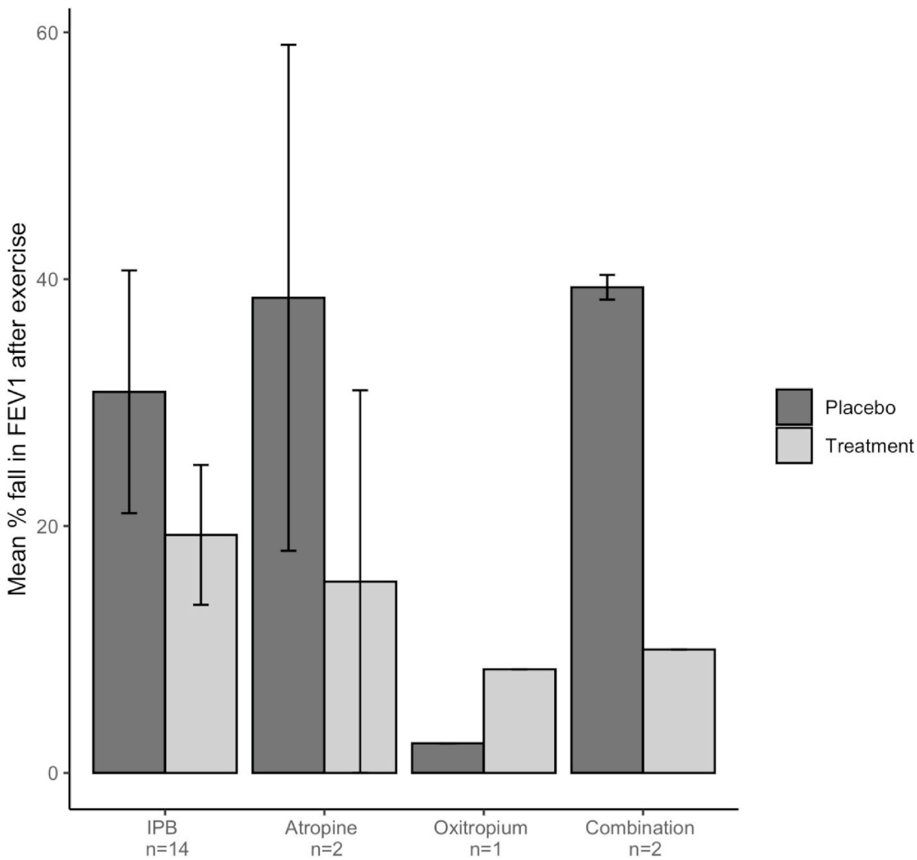

Fig. 2. Mean % fall in FEV_1 after exercise for IPB, Atropine, Oxitropium and Combination of drugs. Note: The number below each treatment indicates the number of studies involved for computation of the FEV_1 fall. The studies included for IPB and the relative sample size of each study were: Bonaventura 2010, n = 21; Boner 1989, n = 12; Chan Yeung 1977, n = 9; Dorward 1982, n = 7; Finnerty 1993, n = 10; Freeman 1992, n = 8; Hartley 1980, n = 10; Knopfli 2004, n = 8; Pasterkamp 1985, n = 6; Poppius 1986, n = 10; Sanguinetti 1986, n = 12; Thomson 1978, n = 13; Yeung 1980, n = 17; Smith 1988, n = 12. The studies included for Atropine and the relative sample size of each study were: Jones 1963, n = 6; Tashkin 1977, n = 15. The study included for Oxitropium and the relative sample size was Taytard 1987, n = 10. The studies included for Combined drugs and the relative sample size of each study were: Thomson 1978, n = 13; Smith 1988, n = 12.

was 73.7% and the difference between Atropine and Placebo was not statistically significant ($p = 0.471$) due to the very limited number of studies. The mean % protection for Combined drugs was 74.6% and the difference between Combined drugs and Placebo was not statistically significant ($p = 0.102$) due to the very limited number of studies.

In conclusion, all the above unmet needs, as well as the potential for an adequate clinical application of the investigated medicines, should prompt additional research based on specifically designed and robust studies in order to more comprehensively understand the exact role of this pharmacological intervention.

Contributions of authors

MB, GC and CB run the search strategy, selected the studies to include in the review, extracted and recorded data. SLG, GV, OU and PP acted as independent reviewers for solving disagreements among rating authors. All authors actively participated in designing the study protocol, drafting the manuscript and revising it critically. All authors approved the final version of the article for submission.

Acknowledgements and Funding

The current work was funded by the Italian Drug Agency (FARM12H2MT) and supported by a ERS/EU RESPIRE2 Marie-Curie Research Fellowship awarded to Matteo Bonini (8051-2015).

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